Clustering I

- Unsupervised learning: basic concepts
- Proximity measures
- K-means: objective, updates, convergence

From dimensionality reduction to clustering

- PCA / KPCA: coordinates preserving variance.
- Laplacian Eigenmaps: preserve local neighborhoods.
- Clustering: discover groups, without known labels.

$$\mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^d \quad \longrightarrow \quad c_i \in \{1, \ldots, k\}$$

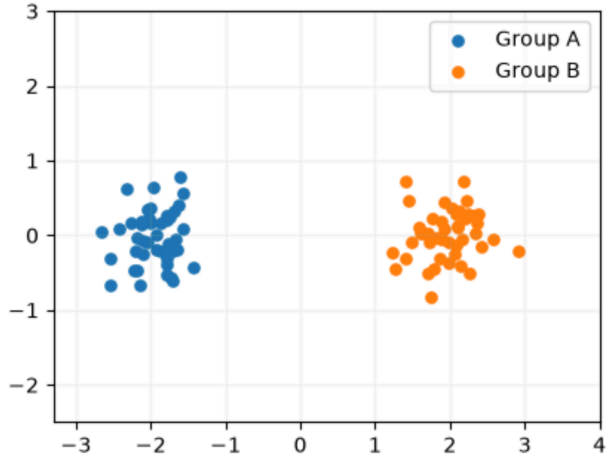
What makes a cluster?

- Compact cloud around a centroid.
- Chain of nearby points: another idea of a group.
- No universal notion of a “correct” partition.

Three geometries, three intuitions

- Compare compact clouds, nested arcs and elongated groups.
- Which shape is easiest to describe with a center?

Geometry (0: clouds · 1: arcs · 2: elongated) 0



Partition, labels, centroids

- Each observation belongs to exactly one cluster.
- The centroid is the mean: not necessarily an observed point.

$$C_r = \{i : c_i = r\}, \qquad n_r = |C_r|, \qquad \boldsymbol{\mu}_r = \frac{1}{n_r} \sum_{i \in C_r} \mathbf{x}_i$$

Three algorithmic families

- K-means: iteratively reassign points to k centers.
- Sequential: decide what to do with the next arriving point.
- Hierarchical: merge currently existing groups, one pair at a time.

Proximity: what does "near" mean?

- Distances: smaller means more similar.
- Similarities: larger means more similar.
- The notion of proximity affects the resulting clusters.

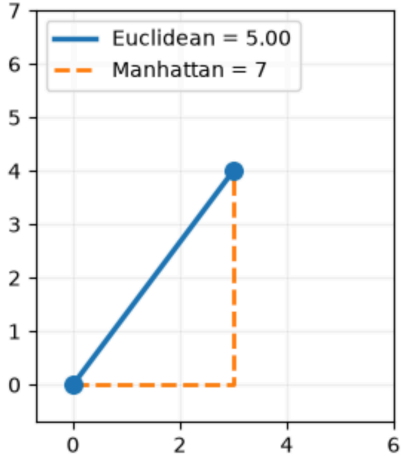
$$d_2(\mathbf{x}_i, \mathbf{x}_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2$$

Euclidean vs Manhattan distance

- Same endpoints; different routes through the plane.
- Move the point to compare d_2, d_1 and d_∞.

$$d_1 = \sum_{\ell=1}^{d} |x_{i\ell} - x_{j\ell}|, \qquad d_\infty = \max_{\ell} |x_{i\ell} - x_{j\ell}|$$

Vertical separation 4



Distances: properties and squared distance

- A metric is nonnegative, symmetric and satisfies the triangle inequality.
- Squared Euclidean distance is useful for k-means but is not generally a metric.

$$\arg\min_r \|\mathbf{x}_i - \boldsymbol{\mu}_r\| = \arg\min_r \|\mathbf{x}_i - \boldsymbol{\mu}_r\|^2$$

Cosine similarity

- Focuses on direction rather than vector length.
- Common for document and high-dimensional vector comparisons.
- Cosine dissimilarity is not automatically a metric.

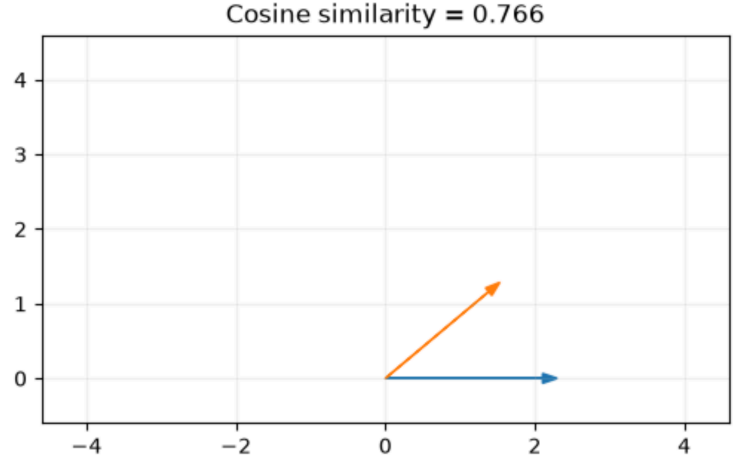
$$s_{\cos}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|}, \qquad \delta_{\cos} = 1 - s_{\cos}$$

Cosine similarity: rotate a vector

- Changing the angle changes the similarity.
- Changing only the length does not.

Angle (degrees) 40

Length 2



Feature scales matter

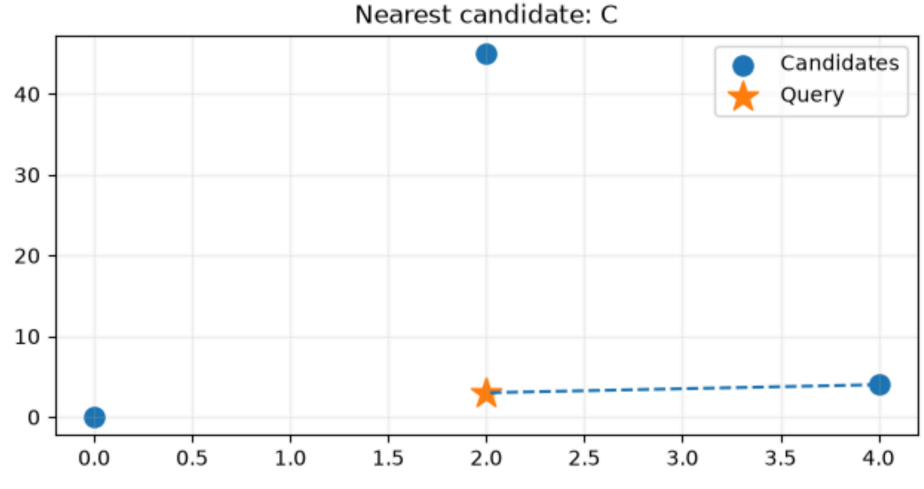
- Large-scale features can dominate Euclidean distances.
- Standardization changes the geometry before clustering.
- Handle constant features separately.

$$\tilde{x}_{i\ell} = \frac{x_{i\ell} - \bar{x}_\ell}{s_\ell} \qquad (s_\ell > 0)$$

Scaling can change the nearest point

- A large feature range dominates before standardization.
- Toggle rescaling to see how the nearest neighbor changes.

Standardize features



Proximity: quick recap

- Physical separation → Euclidean.
- Sum of coordinate deviations → Manhattan.
- Vector orientation → cosine similarity.
- Next: fix squared Euclidean distance and optimize compactness.

K-means: a precise optimization objective

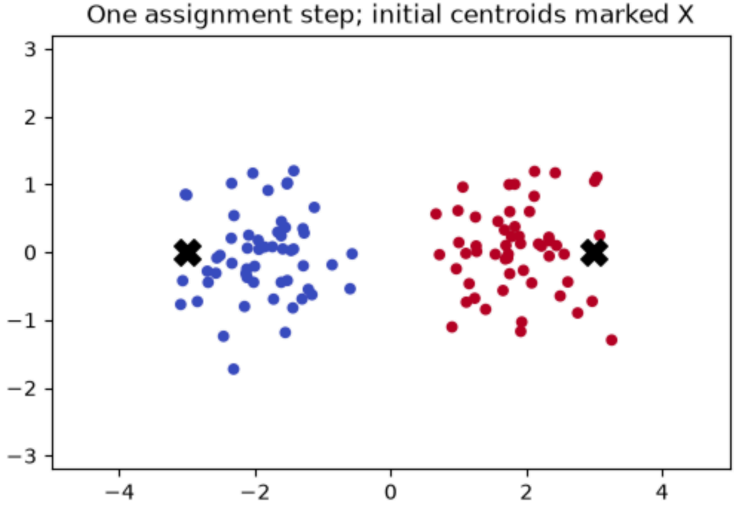
- Input: k clusters and N observations.
- Penalize each squared distance to its assigned centroid.
- Optimize labels and centroids alternately.

$$\boxed{J(\mathbf{c}, \boldsymbol{\mu}) = \sum_{i=1}^{N} \|\mathbf{x}_i - \boldsymbol{\mu}_{c_i}\|^2}$$

K-means: initialization changes the story

- Move the chosen seed configuration.
- Assignments depend on which centroids are initially selected.

Initialization 0



Why is the optimal centroid the mean?

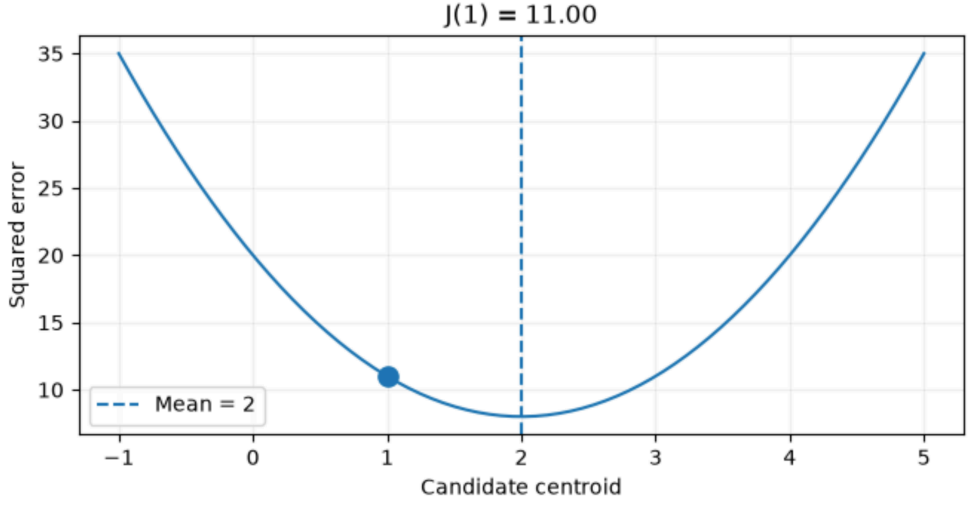
- Hold assignments fixed.
- Minimize within-cluster squared error with respect to the center.
- Derivation on the board → stationary point is the mean.

$$J_r(\boldsymbol{\mu}_r) = \sum_{i \in C_r} \|\mathbf{x}_i - \boldsymbol{\mu}_r\|^2 \quad \Longrightarrow \quad \boxed{\boldsymbol{\mu}_r = \frac{1}{n_r} \sum_{i \in C_r} \mathbf{x}_i}$$

The mean minimizes squared error

- Slide the candidate center.
- The error curve is minimized at the sample mean.

Candidate centroid 1



Why assign to the nearest centroid?

- Now hold centroid positions fixed.
- Each observation affects only its own distance term.
- Minimize independently, one point at a time.

$$\boxed{c_i \leftarrow \underset{1 \leq r \leq k}{\arg\min} \left\| \mathbf{x}_i - \boldsymbol{\mu}_r \right\|^2}$$

Lloyd algorithm: alternating updates

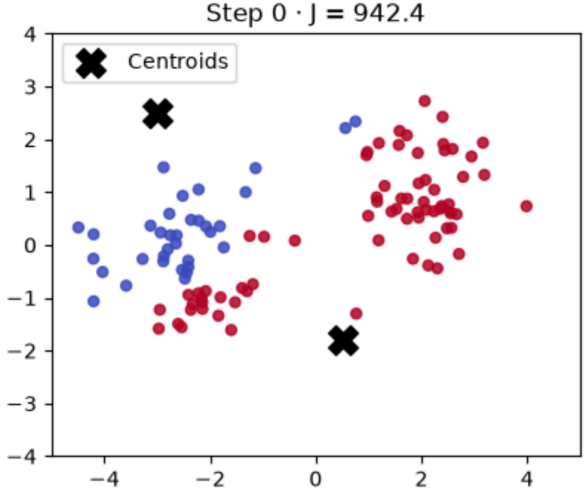
- 1. Initialize k centroids.
- 2. Assignment: nearest centroid.
- 3. Update: arithmetic mean of each nonempty group.
- 4. Repeat until assignments stabilize.

$$c_i \leftarrow \arg\min_r \|\mathbf{x}_i - \boldsymbol{\mu}_r\|^2, \qquad \boldsymbol{\mu}_r \leftarrow \frac{1}{n_r} \sum_{i \in C_r} \mathbf{x}_i$$

K-means animation: assignments and moving centroids

- Advance through initialization, assignment and centroid updates.
- Watch the objective decrease as centers move.

Algorithm step 0



Why does the objective never increase?

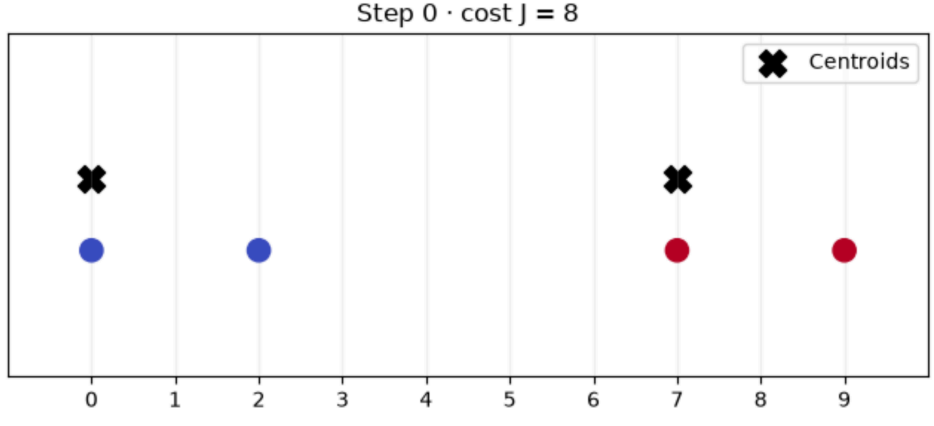
- Assignment step minimizes J with centers fixed.
- Update step minimizes J with labels fixed.
- A decreasing cost does not guarantee a global optimum.

$$J(\mathbf{c}^{(t)}, \boldsymbol{\mu}^{(t)}) \geq J(\mathbf{c}^{(t+1)}, \boldsymbol{\mu}^{(t)}) \geq J(\mathbf{c}^{(t+1)}, \boldsymbol{\mu}^{(t+1)})$$

Worked example: four observations

- Data: $0, 2, 7, 9$; initial centers $0, 7$.
- Assignment → groups $(0, 2)$ and $(7, 9)$; update → means $1, 8$.

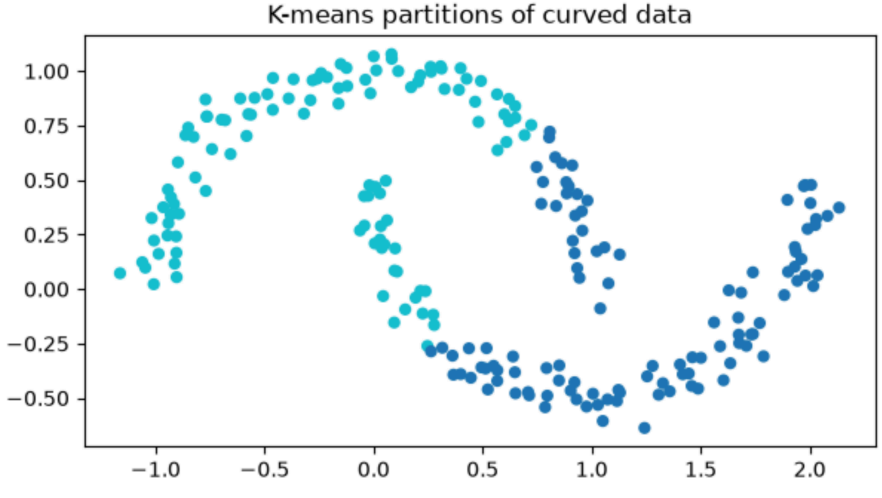
0: initialize · 1: assign · 2: update 0



Geometric limitation: two moons

- K-means creates centroid-based regions, not curved chains.
- The two arcs are not necessarily recovered as separate groups.

Number of clusters k 2



K-means: takeaways

- Fast, intuitive; compact center-based groups.
- Sensitive to scale, outliers and initialization.
- Each iteration reduces (or preserves) squared error.
- No guarantee of global optimum; elongated / curved groups may be difficult.

$$\boxed{J = \sum_i \|\mathbf{x}_i - \boldsymbol{\mu}_{c_i}\|^2 \qquad J^{(t+1)} \leq J^{(t)}}$$